



COLORECTAL CANCER (CRC) DETECTION USING CONVOLUTIONAL NEURAL NETWORK (CNN)

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Abstract

This research focuses on utilizing Convolutional Neural Networks (CNN) to predict the likelihood of colorectal cancer by analyzing medical images of cancerous and non-cancerous tissues. The model demonstrates strong performance with high precision, recall, and accuracy, and employs various techniques such as data augmentation and early stopping to improve generalization and prevent overfitting. The study highlights the potential of machine learning in enhancing diagnostic accuracy and supporting oncologists in making informed treatment decisions.

Keywords: CNN, Colorectal Cancer, Confusion matrix, Medical Imaging.

I. Introduction

Among the various cancers, CRC is ranked among the most common and lethal types worldwide [XIX]. Traditional approaches to colon cancer diagnosis present significant challenges in terms of early identification and accurate diagnosis, making it a prevalent and life-threatening disease [XVI, IX, XII, VII]. To address these challenges and enhance the accuracy and effectiveness of colon cancer diagnosis, advanced technologies such as machine learning and deep learning have been explored in medical image analysis [VIII, IV, XVII]. Artificial intelligence offers solutions to the limitations of traditional approaches by utilizing techniques such as artificial neural networks (ANNs), support vector machine (SVM), fuzzy methods, expert systems, and metaheuristic methods for disease diagnosis using medical images [XIV, III, V, XI, II, I, XX, XXI].

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Convolutional Neural Networks (CNNs) have emerged as a powerful tool in the field of medical imaging, demonstrating significant potential for the automated detection of Colorectal Cancer (CRC). This is primarily because CNNs alter images after receiving, processing, and preserving spatial relationships. Since spatial relationships can be used to read the linkages and interactions between malignant and normal tissue, they are an important consideration in medical picture analysis. CNNs are a deep learning architecture with good image adaptation capabilities that have been developed to address image processing issues. Rather than being directly coupled, the layers in the CNN architecture are connected in different blocks. These blocks transport information similarly to the visual brain, which solves the issues with traditional methods. Additionally, it can learn features from the raw data automatically, avoiding the challenges associated with manual feature extraction. CNNs have fewer parameters than they should since properties are shared. By doing this, the network can facilitate quicker learning and prevent overfitting issues. CNNs are data-hungry deep learning architectures, and millions of parameters can be trained, so they perform well in medical image analysis. The fact that there are many similar advantages paves the way for the use of CNNs in many more areas.

By analyzing various medical image modalities, CNN-based systems can assist in early diagnosis, reduce the workload of pathologists, and enhance diagnostic accuracy by mitigating human error and subjectivity. As such, the objective of the present study is to investigate a comprehensive methodology using CNNs in colorectal cancer detection, demonstrating high accuracy and relevant performance metrics.

The present study is organized as follows: Section II reviews the literature, Section III describes the materials and methods, Section IV presents the results and discussion, and Section V provides the conclusion and future direction.

II. Literature Review

As per the objective of the present study, we have reviewed the articles from several scientific databases such as ScienceDirect, Springer, IEEE Xplore, and ACM Digital Library, by searching the terms “cancer detection using convolutional neural network.”

The following are a few research studies aimed at improving the diagnosis and prognosis of colorectal cancer through innovative methodologies such as automated algorithms, convolutional neural networks (CNNs), and integration of medical data with artificial intelligence (AI) techniques. Each study outlines its aims, advantages, disadvantages, and results, showcasing diverse approaches to addressing the challenges in colorectal cancer diagnosis and treatment.

R. Zhang et al [XXII] developed an automated algorithm for detecting and categorizing hyperplastic and adenomatous colorectal polyps during colonoscopy. Their proposed algorithm demonstrated comparable precision but a superior recall rate and accuracy compared to visual inspection by endoscopists.

J S Nisha et al [XIII] proposed a Dual-Path Convolutional Neural Network (DP-CNN) to automatically detect intestinal polyps from colonoscopy images. Their

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model achieved high accuracy in detecting polyps, with recall rates of 99.20% and 92.85%, precision rates of 100% and 89.81%, F1-Scores of 99.60% and 91.00%, and F2-Scores of 99.83% and 89.91% on the CVC ColonDB and ETIS-Larib databases, respectively.

L Nanni et al [XV] innovated ensembles integrating convolutional neural networks (CNNs) and transformers for semantic segmentation to validate the efficacy of merging diverse models, training methodologies, and optimization techniques to forge more potent ensembles. It was a novel strategy of averaging intermediate predictions that significantly contributes to mitigating overfitting and refining model contributions, underscoring its pivotal role in the ensembles' success.

Y Du et al [VI] demonstrate the viability of integrating serum Raman spectroscopy with a convolutional neural network (CNN) model for the diagnosis of four cancer types: gastric, colon, rectal, and lung cancer. It achieved a notable 94.5% accuracy in diagnosing multiple cancer types.

A Karthikeyan et al [X] proposed a model, employing CNN and a Ranking algorithm, that demonstrates superior performance in colorectal cancer diagnosis compared to existing methods, as indicated by higher Recall, Precision, and Accuracy metrics.

The present study aims to predict the possible incidence of colorectal cancer based on clinical and demographic data. This is possible through the application of CNN, trying to improve diagnostic precision and possibly leading to better targeted screening and care strategies. A method trains algorithms from data sets for the identification of critical CRC indicators: genetic markers, lifestyle, and symptom profiles; for later stages, it results in better chances of the early detection and prognosis of CRC patients.

III. Materials and Methods

The dataset [VIII] for this study is made up of medical images divided into two categories: Colon_Aca, which includes images of cancerous colon tissue, and Colon_N, representing images of benign colon tissue. A function is created to systematically load and label these images, storing their file paths and corresponding labels in a Pandas DataFrame for easy access. To ensure consistency, the images were resized and normalized to standard dimensions. The dataset is then split into training (64%), validation (16%), and test (20%) sets using stratified sampling with `train_test_split` from `sklearn.model_selection`, which helps maintain a balanced representation of each class.

To classify the medical images, a Convolutional Neural Network (CNN) model using TensorFlow and Keras is implemented. This CNN model consists of several important layers: convolutional layers that extract spatial features from the input images, max pooling layers that reduce dimensionality while keeping essential features, a flatten layer that transforms feature maps into a one-dimensional vector, fully connected layers that learn complex representations for classification, and a Softmax output layer that predicts the likelihood of an image being either cancerous or benign.

For the loss function, Categorical Crossentropy, ideal for multi-class classification, is used, and then the Adam optimizer for adjusting the learning rate adaptively. Performance metrics like accuracy, precision, recall, and F1-score are closely monitored. Data augmentation technique in terms of brightness, rotation, and flipping is also applied. To avoid over-fitting, the training is halted when the validation loss stops improving.

Input images were preprocessed by resizing them to 224×224 pixels and normalizing the pixel values. These images were then converted to NumPy arrays and expanded to fit the model's expected input shape. Finally, the model generated predictions, classifying the images as either cancerous or benign.

IV. Results and Discussion

Our CNN model was tested on images of colon tissue, divided into two groups they are Benign (normal) and Cancerous (adenocarcinoma). The model reached an overall accuracy of about 89%, showing that it can be a useful tool for helping doctors detect colorectal cancer. At the same time, the results also point to a few challenges that need improvement.

a. Training and Validation

When training the model, accuracy on the training data kept improving. However, accuracy on the validation data stopped improving after a point and even started to drop slightly. The training loss (errors) went down, but the validation loss began to rise. This shows the model was overfitting. It was learning the model training data too well, but not generalizing as strongly to new, unseen data. Early stopping helped, but more measures are needed to prevent overfitting.

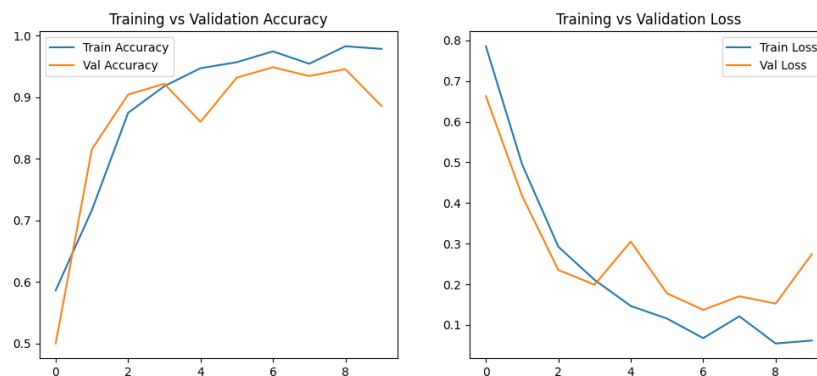


Fig 1. Training and validation graph

b. Classification Results

The model's performance can be better understood through Precision, recall, and F1 score.

- **Precision**-Out of all the cases the model predicted as positive, how many are actually correct?

- **Recall**-Out of all the actual positive cases, how many can the model correctly find?
- **F1-Score**- A balanced score that combines precision and recall into one number. It checks overall performance.

Benign tissue (normal tissue):

- **Precision = 0.96** - When the model predicted “benign”, it was correct 96% of the time.
- **Recall=0.82** - It correctly found 82% benign cases, but it still missed some, labeling them as cancerous.
- **F1-score=0.88** - A balanced measure of 88% accuracy showing good performance.

Cancerous tissue:

- **Precision=0.84** - The model finds 84% cancerous, but some benign samples were wrongly marked as cancerous.
- **Recall=0.96** – The model caught nearly all cancerous cases, which is 96%. It is very important in diagnosis.
- **F1-score=0.90** – The model shows an overall 90% accuracy for detecting cancer.

The high recall for cancer means the model rarely misses actual cancer cases, which is vital. However, the lower recall for benign tissue means it sometimes raises false alarms.

Confusion Matrix

The Confusion Matrix confirmed these patterns that the model is very strong in detecting cancer, but sometimes mistakes healthy tissue for cancer. In a clinical setting, this could lead to extra, unnecessary tests. While this is less harmful than missing a cancer case, it still needs to be reduced for the model to be practical in real medical use.

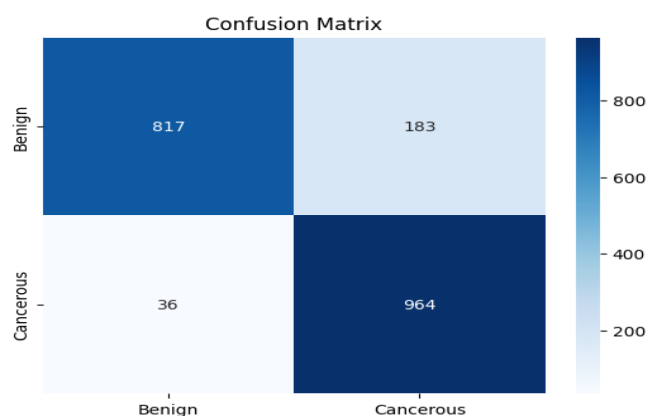


Fig 2. Confusion Matrix

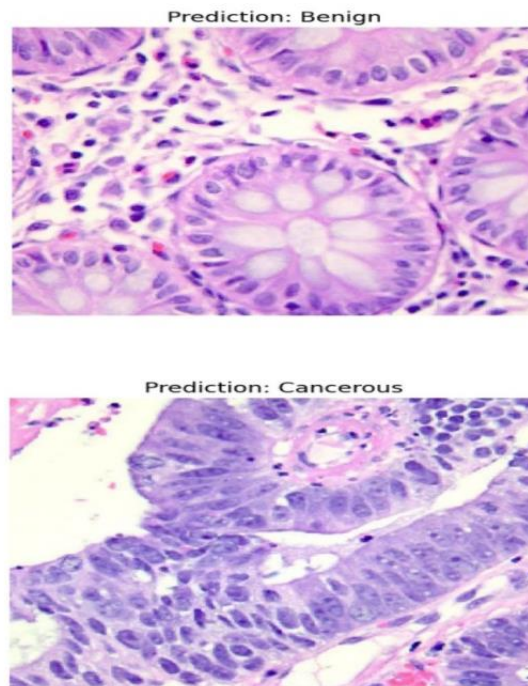


Fig 3. Picture that shows two different cells

These findings show that CNNs can support cancer diagnosis by quickly analyzing tissue images with good accuracy. The model is especially reliable at catching cancer cases, which is the main goal. However, false positives (normal tissue marked as cancer) needed to be reduced. To improve this, techniques like dropout, weight regularization, more varied data augmentation, or combining CNNs with other deep learning models could be used.

V. Conclusion

The paper presents a comprehensive methodology for using CNNs in colorectal cancer detection, demonstrating high accuracy and relevant performance metrics. This research addresses a critical area in medical diagnostics, contributing valuable insights into early cancer detection.

The confusion matrix, as well as the classification report, clearly indicates the overall good accuracy of the model, revealing an excellent ability to identify cancerous cases. More techniques, including data augmentation, advanced feature engineering, and the incorporation of domain knowledge into the model, might help further enhance its performance and reliability. Overall, the present study demonstrates significant potential for enhancing early detection and diagnosis of colon cancer, thereby facilitating timely intervention and improving patient prognosis.

It is also possible to further optimize the decision-making process within the model, particularly borderline ones, so as not to contribute to false negatives. Thus, this way, the model can be improved to provide more accurate and reliable predictions, thereby

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helping healthcare professionals make the right decisions, leading to better patient outcomes. Nevertheless, it's crucial to note that AI models are tools, and their predictions should always be interpreted in the context of the opinion of experts and clinical background.

Conflict of Interest

The author declares that there is no conflict of interest regarding this paper.

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